

GAMERHUB Roadmap

Phase 0 — Project Foundation

1. Spring Boot Architecture
2. Maven
3. Dependency Management
4. application.yml
5. Profiles
6. Configuration
7. Lombok
8. DevTools
9. Validation
10. Logging

Phase 1 — Database Module

1. Spring Data JPA ,Hibernate
2. MySQL ,Flyway
3. Entity ,@Table
4. @Column,Relationships
5. OneToOne ,OneToMany
6. ManyToMany,Cascade
7. FetchType ,Lazy Loading
8. JPQL ,Native Query ,Repository
9. Pagination ,Sorting ,Auditing

Phase 2 — DTO + Validation Module

1. DTO ,Request DTO
2. Response DTO
3. Validation
4. @NotBlank ,@Email ,@Size
5. Custom Validation
6. Global Validation

Phase 3 — Mapper Module

1. Manual Mapper
2. MapStruct
3. Nested Mapping
4. Custom Mapping
5. Ignore Fields
6. Expression

Phase 4 — Exception Handling Module

1. Custom Exception
2. Global Exception
3. @ControllerAdvice
4. Error Response
5. Business Exception
6. Validation Exception

Phase 5 — Authentication Module

1. Password Encoding
2. BCrypt
3. AuthenticationManager
4. UserDetails
5. UserDetailsService
6. SecurityFilterChain
7. AuthenticationProvider

Phase 6 — JWT Module

1. JWT
2. Access Token
3. Refresh Token ,JwtFilter
4. Claims ,Expiration
5. AuthenticationEntryPoint
6. Security Context

Phase 7 — OAuth2 Module

1. OAuth2
2. Google Login
3. OAuth2User
4. CustomOAuth2UserService
5. Success Handler
6. Failure Handler

Phase 8 — Authorization Module

1. Roles
2. Permissions
3. Authorities
4. @PreAuthorize
5. Method Security ,HasRole
6. HasAuthority

Phase 9 — User Module

1. Profile
2. Upload Avatar
3. Follow Players
4. Search Users

Phase 10 — Team Module

1. Create Team
2. Join Team
3. Leave Team
4. Invite Members
5. Captain
6. Owner

Phase 11 — Tournament Module

1. Create Tournament
2. Join Tournament
3. Manage Players
4. Tournament Status

Phase 12 — File Upload Module

1. MultipartFile
2. Validation
3. Storage
4. Cloudinary أو MinIO (لاحقاً)

Phase 13 — Documentation Module

1. Swagger
2. OpenAPI
3. API Versioning
4. Response Examples

Phase 14 — Testing Module

1. JUnit
2. Mockito
3. Integration Test
4. TestContainers

Phase 15 — Docker Module

1. Dockerfile
2. Docker Compose
3. MySQL Container
4. Environment Variables

Phase 16 — Deployment Module

1. Production Profile
2. Render
3. Railway
4. AWS EC2
5. Nginx
6. HTTPS
7. CI/CD

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├─ config

├─ controller

├─ service

| └─ impl

├─ repository

├─ entity

├─ dto

| └─ request

| └─ response

├─ mapper

├─ exception

├─ security

├─ validation

├─ util

├─ constant